

Customer Shopping Behavior Analysis

Business Problem Statement

1. Business Problem

A retail company wants to better understand its customers' shopping behavior and purchasing patterns. The company aims to identify factors associated with customer spending, product preferences, discounts, subscriptions, and repeat purchases.

2. Project Objective

The objective of this project is to analyze customer shopping data and identify meaningful patterns that can help understand customer behavior and support data-driven business decisions.

3. Key Areas of Analysis

- Customer demographics and purchasing behavior
- Product preferences and sales
- Customer spending and revenue
- Discount usage
- Subscription behavior
- Customer segmentation
- Repeat purchasing patterns
- Product ratings and shipping preferences

4. Project Approach

Stage	Tools	Activities
Data Preparation	Python / Pandas	Explore, clean, transform, and prepare the dataset for analysis.
Data Analysis	SQL / PostgreSQL	Analyze customers, purchases, revenue, products, discounts, and customer segments.
Visualization	Power BI	Build an interactive dashboard to present important customer and purchasing insights.

5. Expected Outcome

The analysis aims to identify useful customer and purchasing patterns and present them through SQL-based analysis and an interactive Power BI dashboard.

Tools: Python (Pandas) • PostgreSQL / SQL • Excel • Power BI